



Graduate Education and
Postdoctoral Affairs



Isoform-Aware Variant Annotation

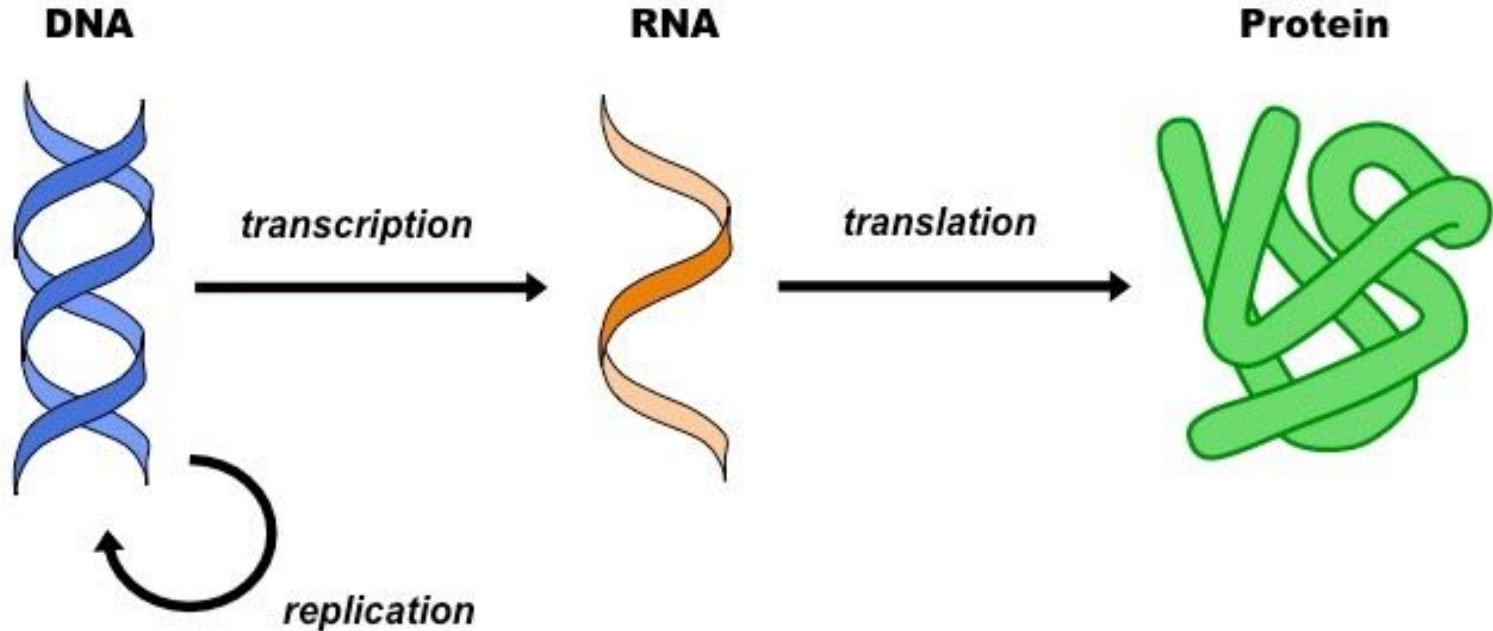
Mapping genetic variants across a gene's full
transcript landscape

Fabrizio Malatesta - Willsey / Wang Lab

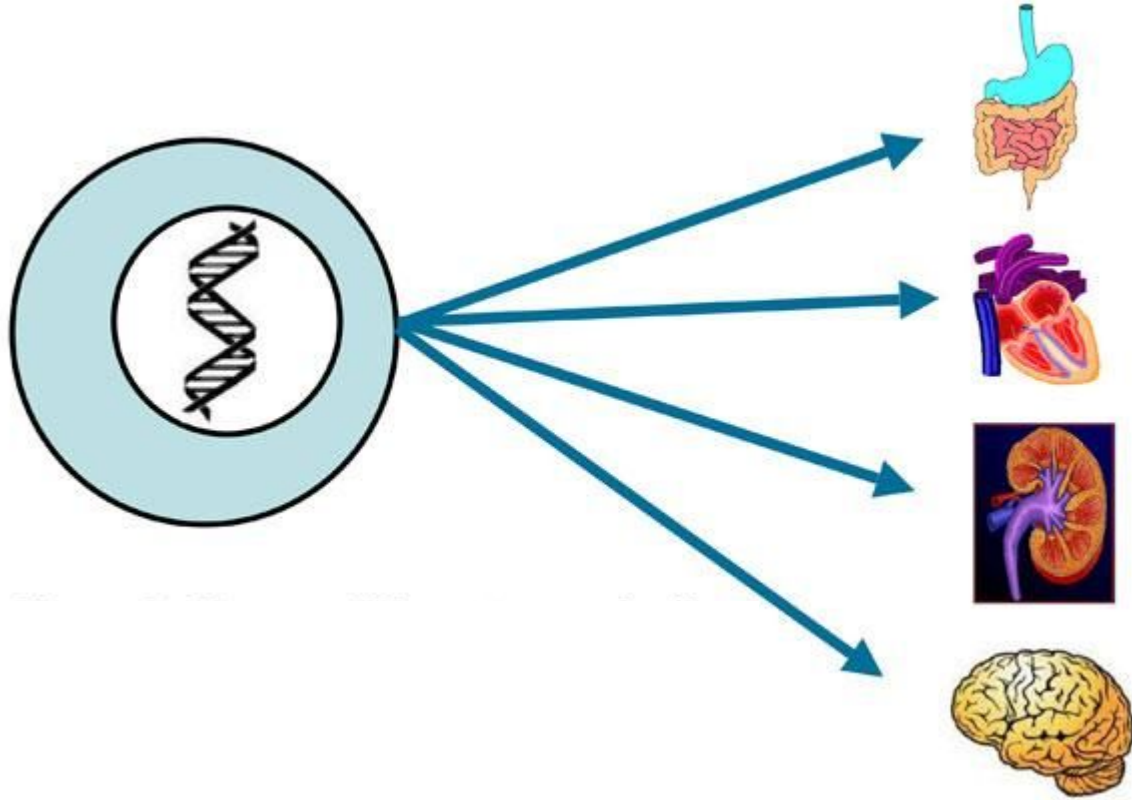
Mentors: Helen Willsey, Ph.D., Belinda Wang, M.D., Ph.D., Matt N Tran, Ph.D.

UCSF Summer Research Training Program · 2026

From blueprints to buildings: The central dogma

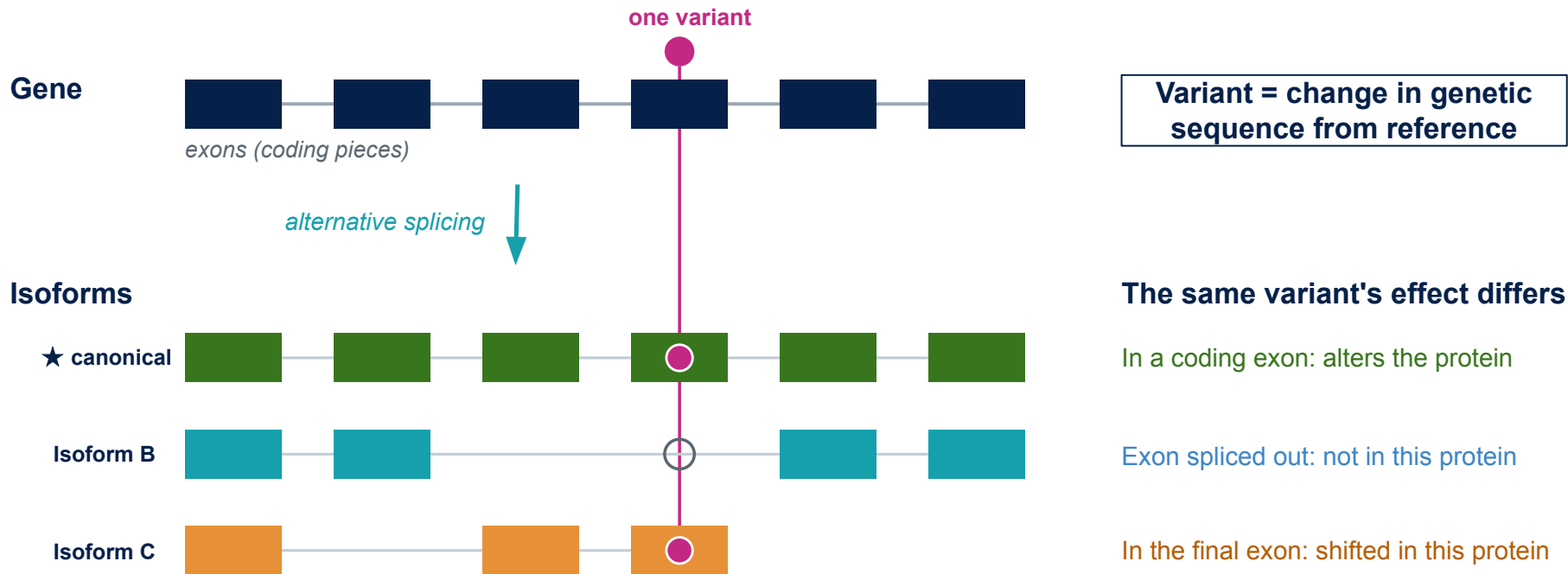


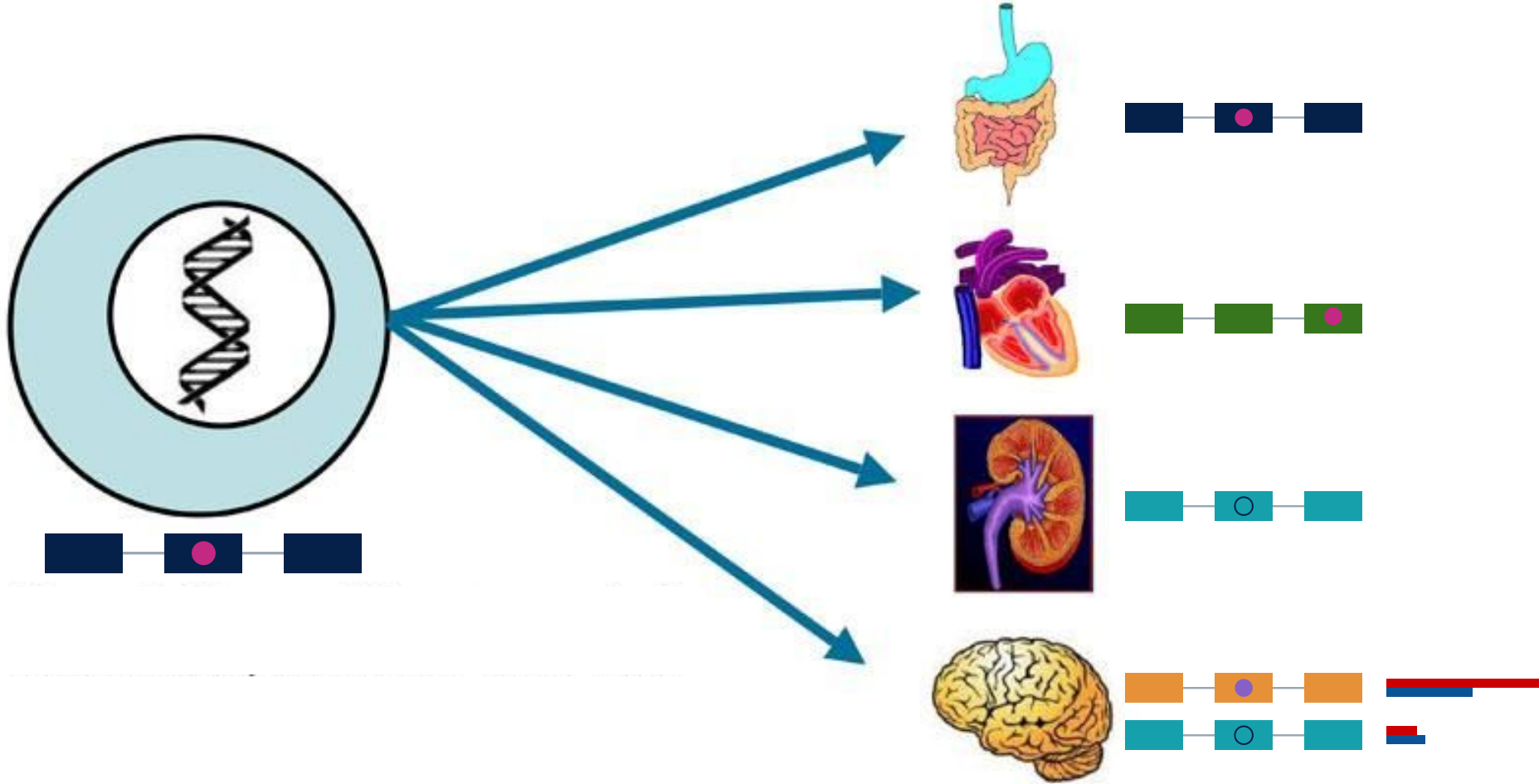
Cells: same blueprints, different buildings?



One gene $\rightarrow$ many isoforms

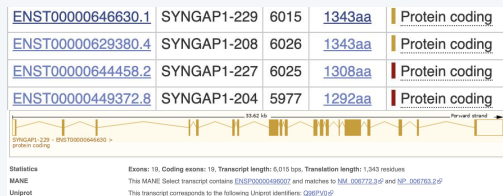
One variant $\rightarrow$ many consequences





No existing tool shows every variant on every isoform. Current tools allow us to:

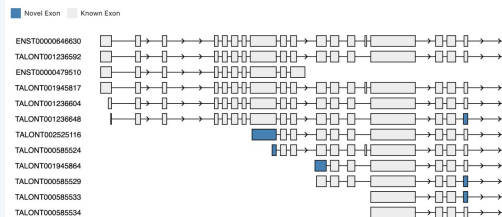
See annotated “reference” isoforms with more detail



misses novel isoforms

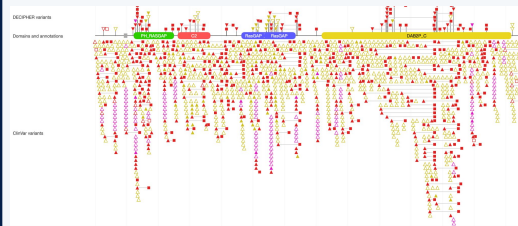
Explore experimentally described novel isoforms

Chr6 — SYNGAP1



limited detail: no variant or domain context

Observe all published patient variants

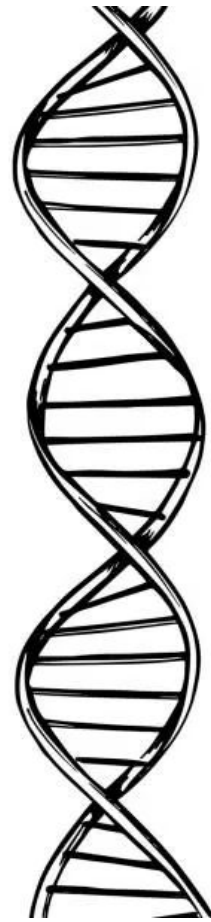


but mapped only to the canonical isoform

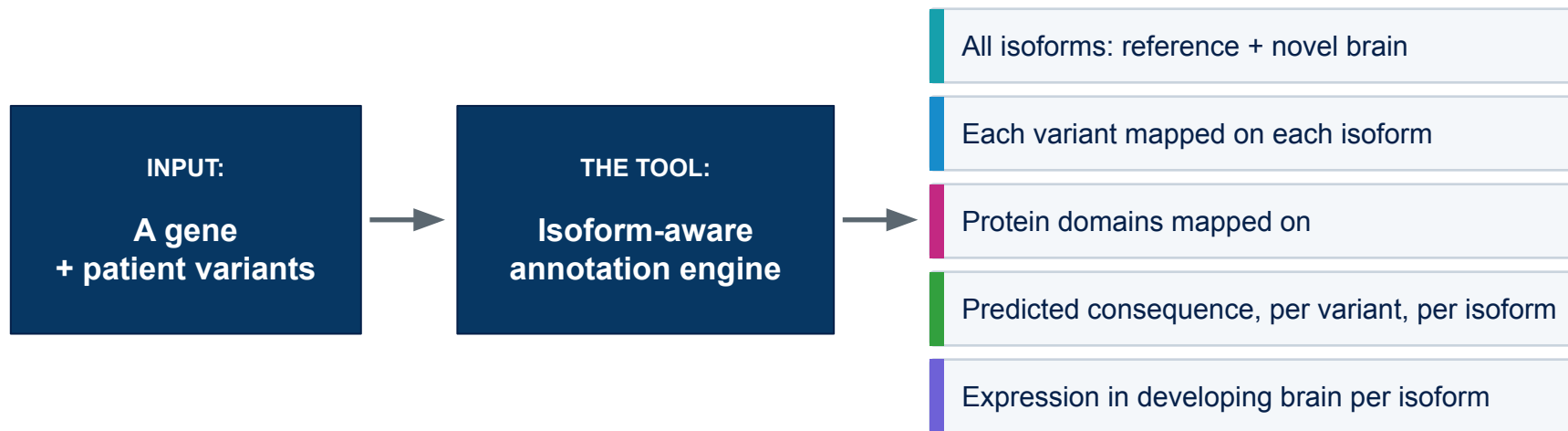
What we wanted: each variant mapped onto each isoform: reference AND experimentally-found brain forms.

Our hypothesis

Integrating patient variants with experimentally-described novel transcript isoforms can allow us to identify consequential isoforms beyond the canonical transcript.



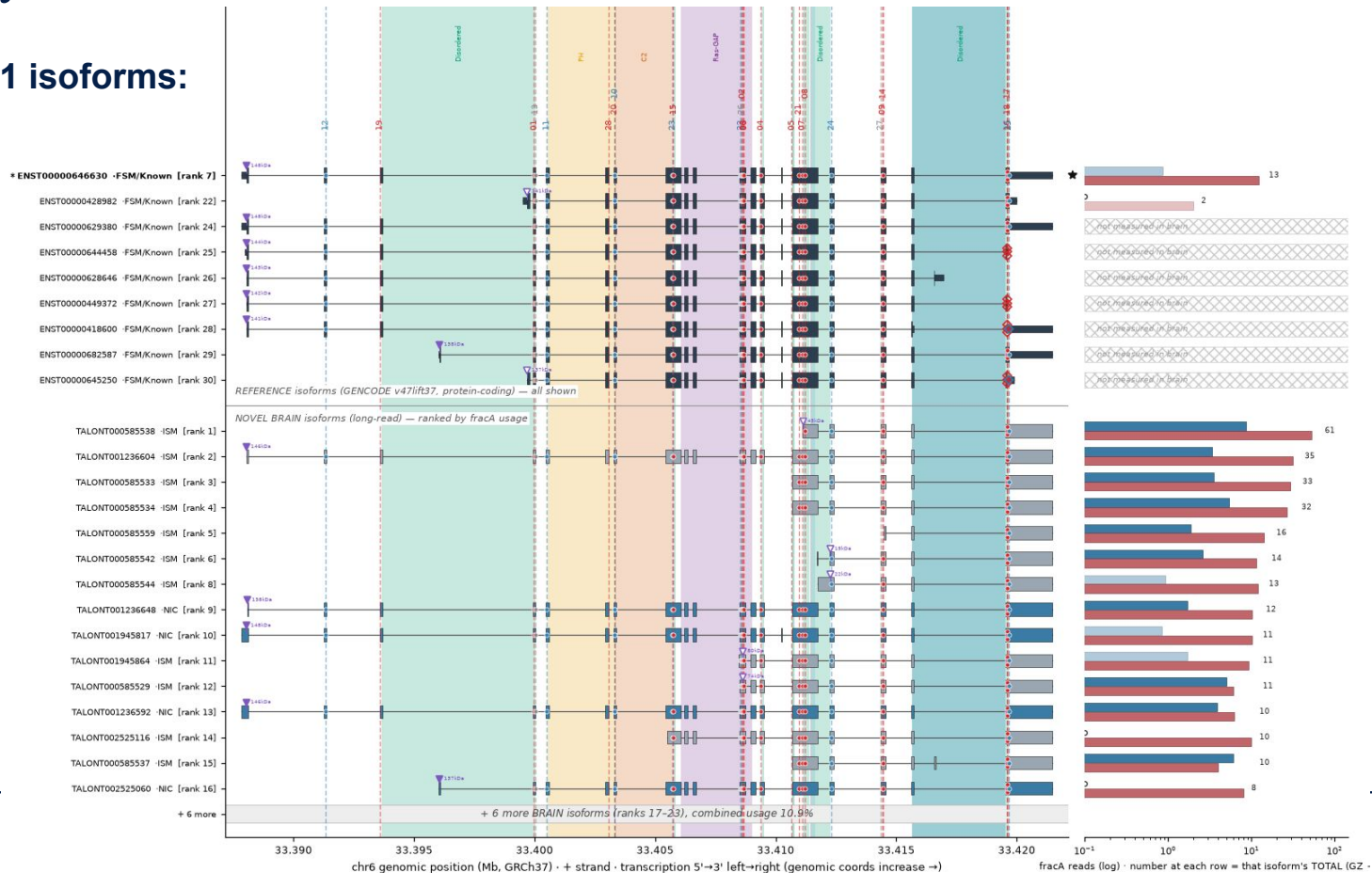
I created a tool to map any variant across every isoform for any gene



*Built on: reference transcripts (GENCODE v47lift37) - variant consequences (Ensembl VEP) ·
brain long-read RNA (Patowary et al. 2024) - protein domains (UniProt)*

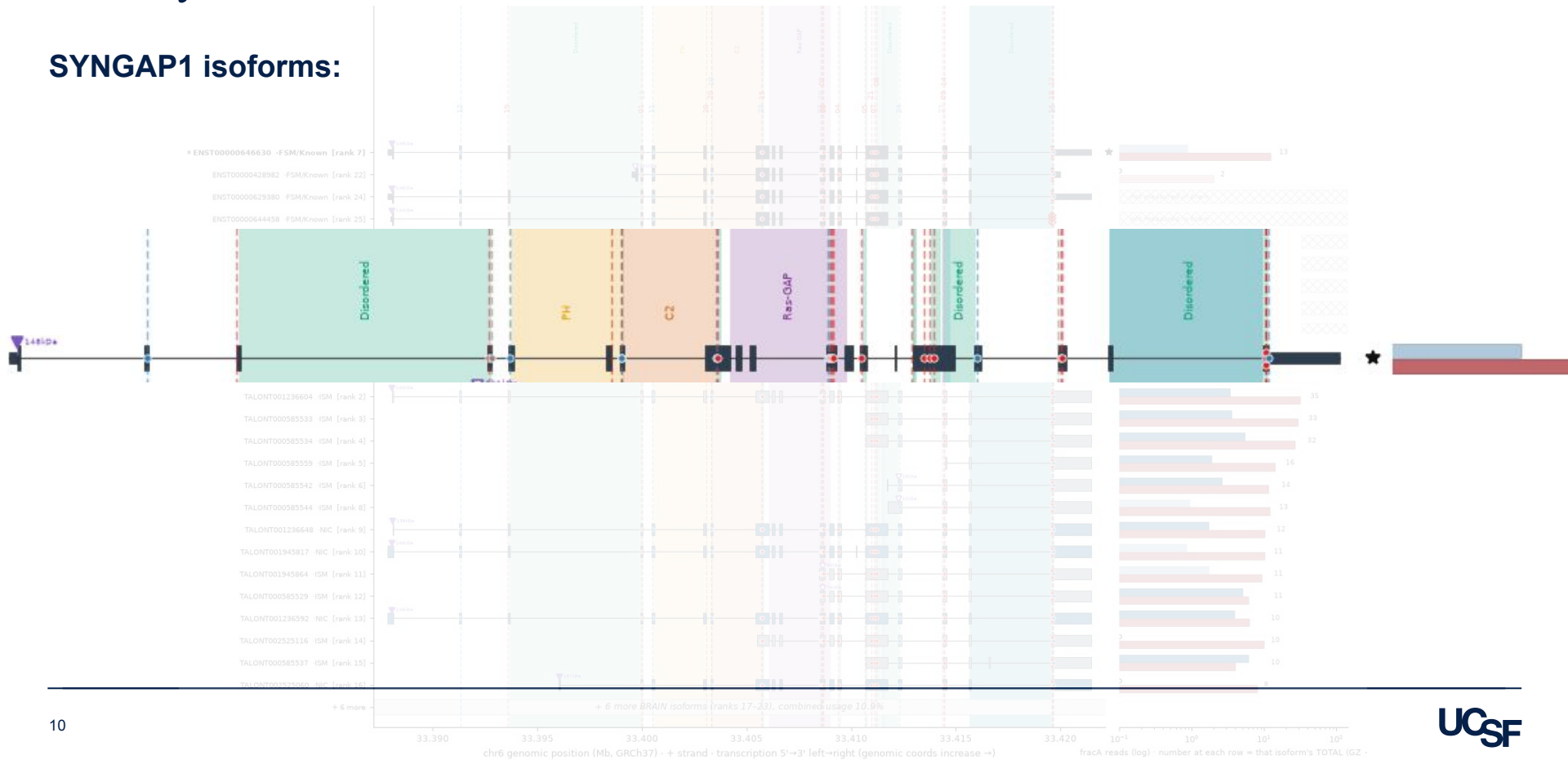
Every isoform in one view, with variants on each

SYNGAP1 isoforms:



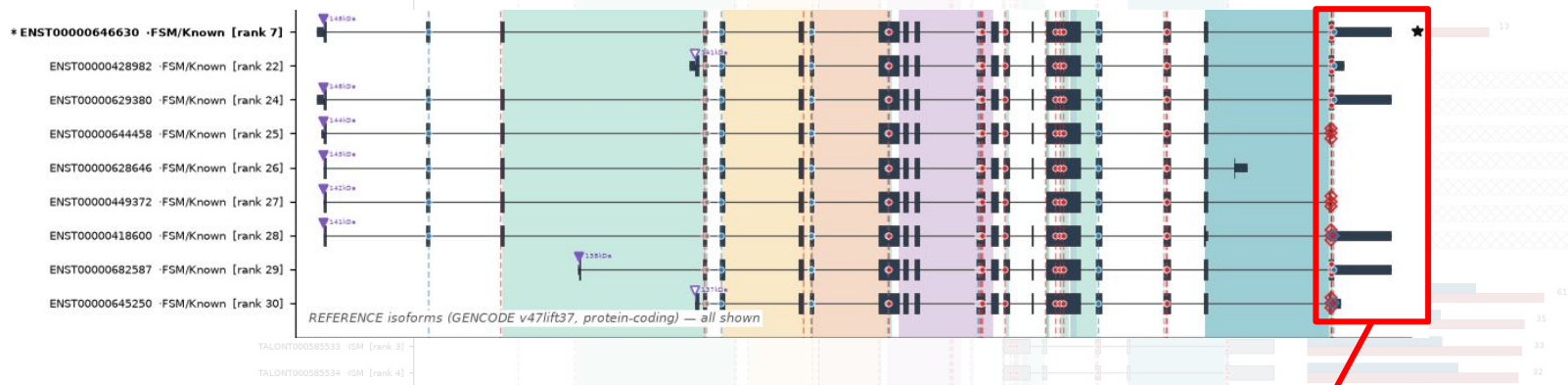
Every isoform in one view, with variants on each

SYNGAP1 isoforms:



Variant Consequence is Isoform-Dependent

SYNGAP1 reference isoforms:



★ ENST00000646630 ·FSM/Known [rank 7]

ENST00000428982 ·FSM/Known [rank 22]

ENST00000629380 ·FSM/Known [rank 24]

ENST00000644458 ·FSM/Known [rank 25]

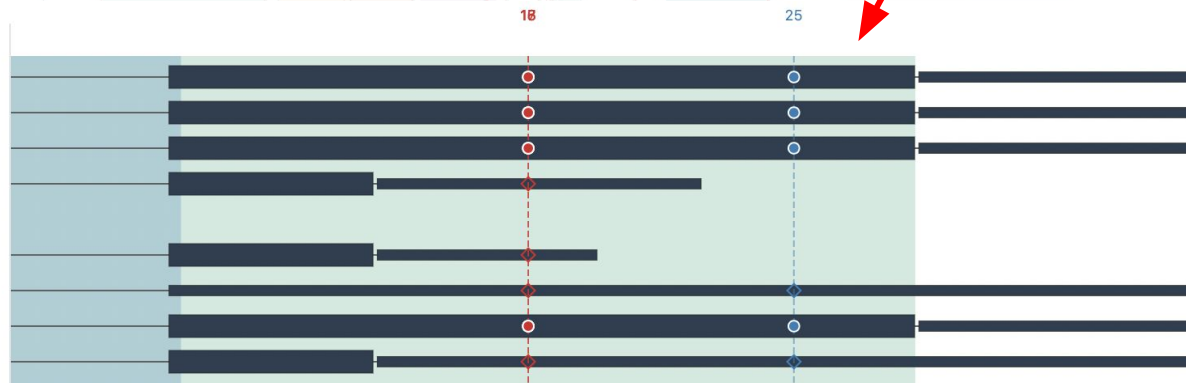
ENST00000628646 ·FSM/Known [rank 26]

ENST00000449372 ·FSM/Known [rank 27]

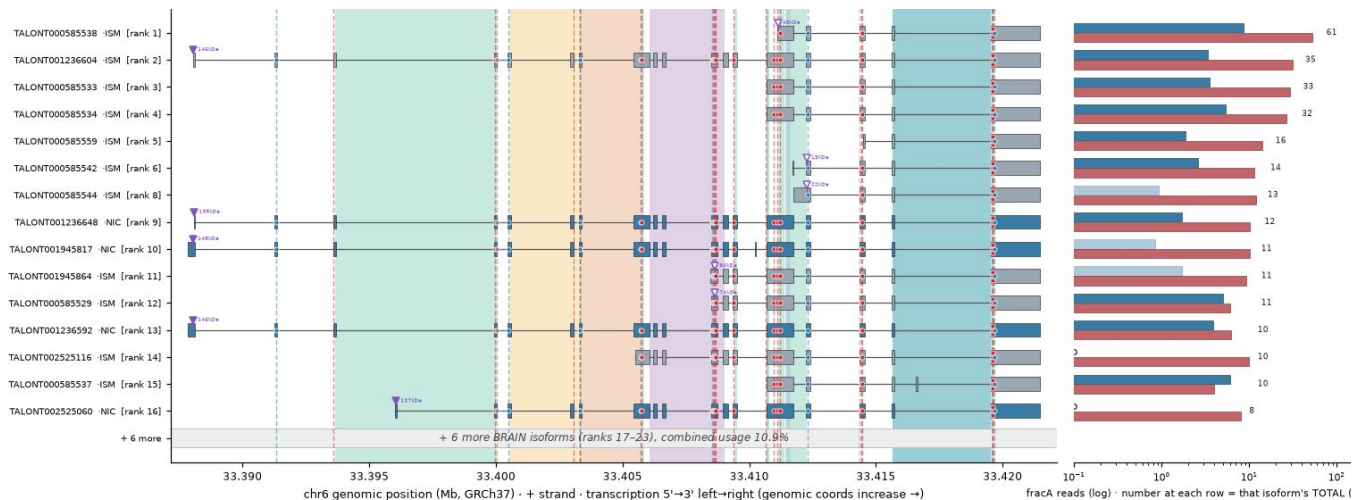
ENST00000418600 ·FSM/Known [rank 28]

ENST00000682587 ·FSM/Known [rank 29]

ENST00000645250 ·FSM/Known [rank 30]



The tool highlighted SYNGAP1 isoforms missing from the reference



What the tool showed

Running it on SYNGAP1 highlighted novel isoforms: real transcripts absent from the reference annotation, that patient variants map onto.

Exactly the isoforms standard tools ignore

Could these isoforms explain an unexplained protein band?



Connecting to the bench

A Western blot separates a protein by size; each band is a protein of a given weight.

Could one of these novel isoforms be the source of that band?

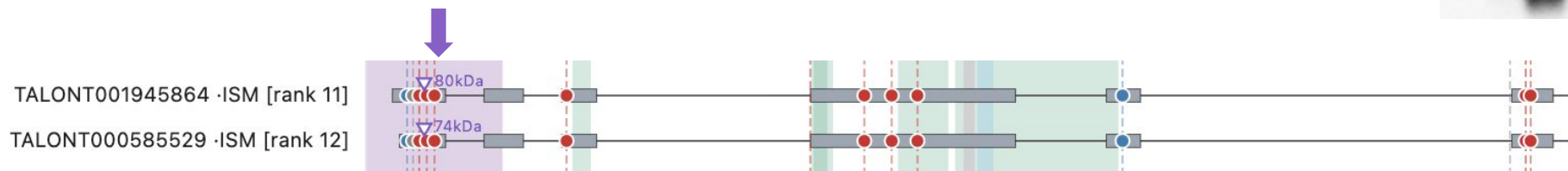
Gap: The tool only describes **transcription**, not **translation**

Predicted translation flagged two candidate isoforms

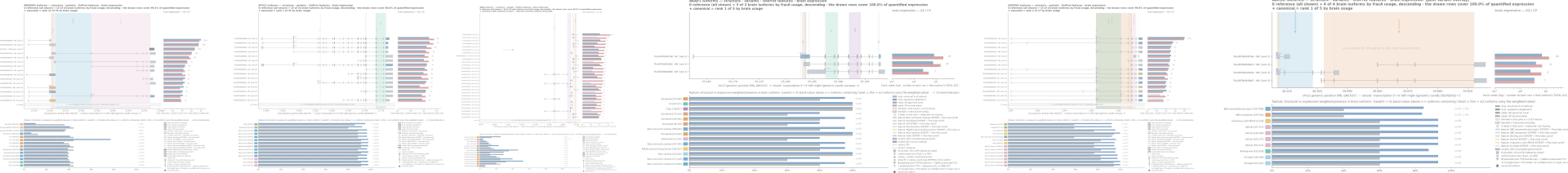
I added predicted translation for every isoform

The tool now predicts and analyzes every possible protein product across each isoform, returning likely products and molecular weights, in kDa, the unit a Western blot measures.

Two of the novel isoforms are predicted to make products of sizes consistent with the unexplained band:



Next step is wet-lab work to test whether these isoforms are indeed transcribed and translated.



Beyond SYNGAP1: a tool for any gene

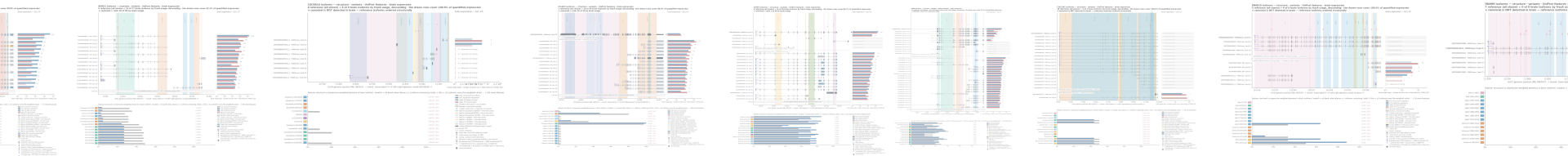
SYNGAP1 became a deep case study, but the tool is general.

For any gene + variant list, it will:

- Map every variant across every isoform
- Include experimentally-found novel brain isoforms
- Predict each isoform's protein product & size
- Read consequences in the context of domains

The goal

Keep improving features & usability, then release it for public use by researchers studying the genetic basis of disease.



What this can't tell us

- Brain isoforms come from developing (mid-gestation) neocortex; prenatal cortex, not adult brain or other tissues
- Isoforms may reflect incomplete transcripts rather than distinct products
- Published patient variants are ascertainment-biased, reported largely because they were interpretable on the canonical transcript, and clinical databases underrepresent non-European ancestry
- Translation is predicted: ORFs are computational, and molecular weight from sequence ignores post-translational modification
- Research-use hypothesis generation, not clinical variant classification

Directly useful to labs interpreting variants in genes with complex brain isoform structure; before it could inform patient care it would need functional validation and integration with clinical classification frameworks.

Conclusions & next steps

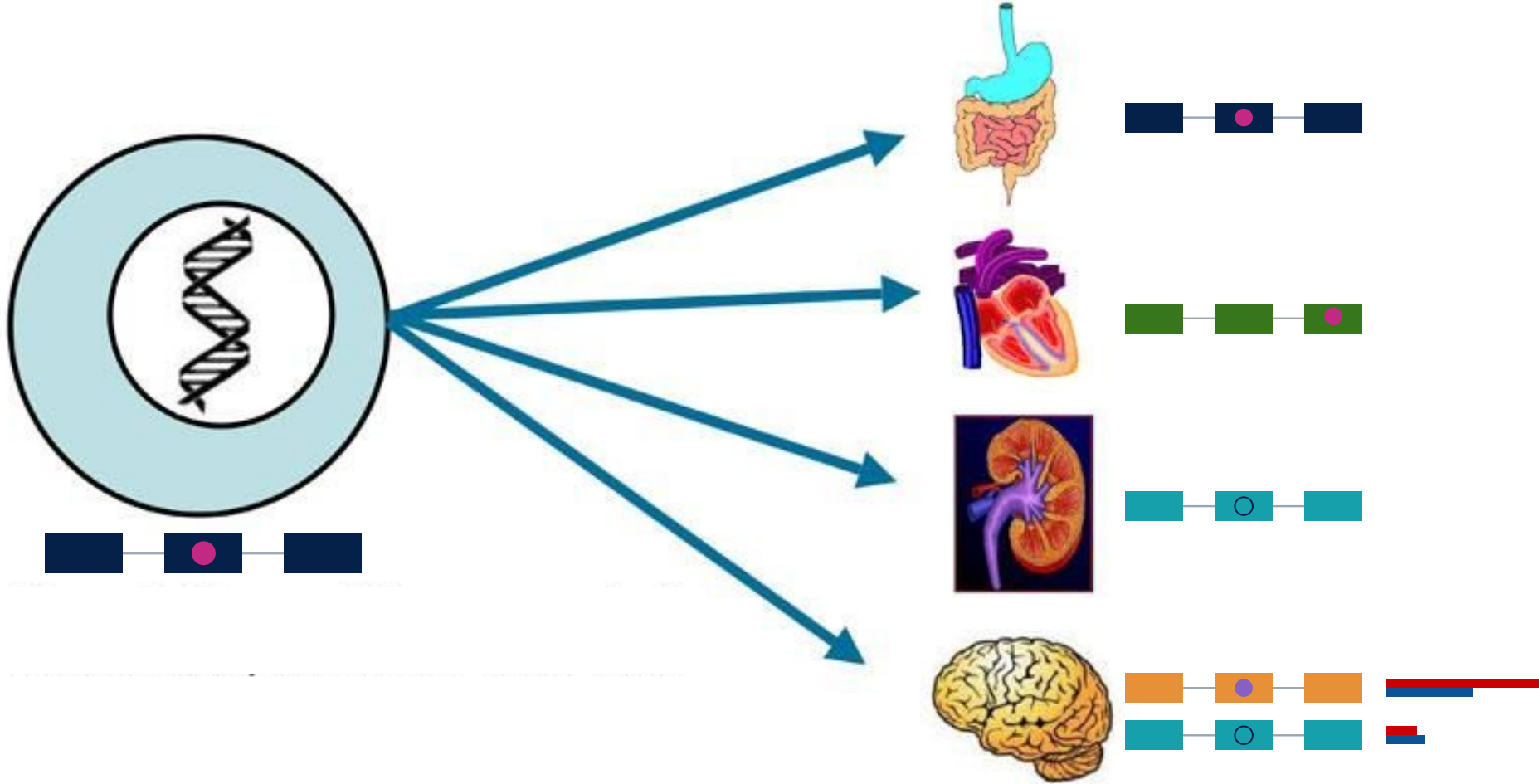
Conclusions:

- We addressed the need for a tool that integrates variant mapping with transcript isoform-level resolution
- In SYNGAP1 it surfaced isoforms missing from the reference that patient variants map onto, and flagged two whose predicted products match a previously unexplained band
- The approach generalizes to any gene + variant list

Next steps:

- Wet-lab validation of the SYNGAP1 protein
- Apply across more disease genes
- Improve features & usability → public release

Why it matters: Reading variants against the RIGHT isoform, not just the canonical, offers a more comprehensive basis for interpreting variants and develops our understanding of genetic etiology, not just for autism, but for any condition



Acknowledgements

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Thank you - Questions?